

Root-Cause Analysis & S&OP Recovery Plan

Product Family: EcoClean Pods

Review Horizon: Weeks 1–8

Planning Horizon: Weeks 9–21 (13 Wks)

Scope Level: Retail & DTC Channels

01 EXECUTIVE SCOPE & SCOPE DEFINITIONS

This report evaluates the operational performance and financial variance across the **EcoClean Pods** product family during the **8-week historical review window (Weeks 1–8)** to recalibrate the forward-looking **13-week planning horizon (Weeks 9–21)**.

Core Scope Definitions

- Actual Demand Definition:** Defined as *Actual Customer Orders* (93,000 units), representing unconstrained customer interest, rather than Net Shipments (85,500 units).
- Cost Baseline:** Planned operational spend of **\$740,000** vs. Actual Spend of **\$864,000**, creating a **\$124,000 unfavorable cost variance (+16.8%)**.
- Analysis Level:** Product family, channel (Retail & DTC), and operational nodes (Plant & DC).

Strategic Value to Executive Leadership

This scope directly answers leadership questions by isolating where customer demand diverged from operational plans, tracing how promotional shifts and supplier disruptions degraded DC service levels, and connecting operational breakdowns directly to financial cost variances.

02 FORECAST ACCURACY & SEGMENTATION EVALUATION

Forecast and actual customer orders were aligned weekly at the product family level. Scale-sensitive (**MAE**) and scale-normalized (**MAPE** / **WAPE**) metrics were calculated using standard formulas: $MAE = (1/n) \sum |Actual_t - Forecast_t|$ | $WAPE = (\sum |Actual_t - Forecast_t|) / \sum Actual_t$

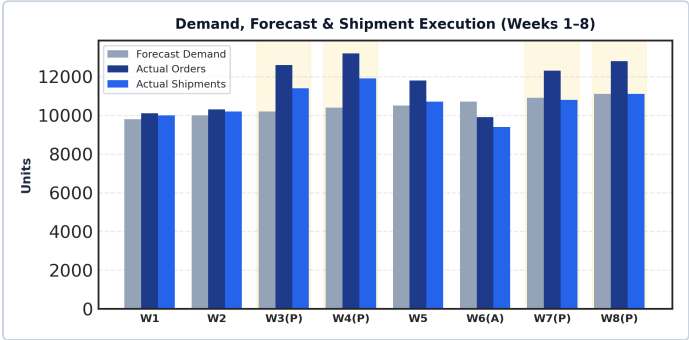
Week	Segment	Forecast	Actual Orders	Shipments	Abs Error	MAPE	Backorders	Operational Notes
Week 1	NORMAL	9,800	10,100	10,000	300	2.97%	100	Normal baseline demand
Week 2	NORMAL	10,000	10,300	10,200	300	2.91%	100	Normal baseline demand
Week 3	PROMOTION	10,200	12,600	11,400	2,400	19.05%	1,200	Retail promo pulled forward +1 week
Week 4	PROMOTION	10,400	13,200	11,900	2,800	21.21%	1,300	Promotion continues; supplier delayed
Week 5	CONSTRAINT	10,500	11,800	10,700	1,300	11.02%	1,100	Backorders overflow into Week 5
Week 6	ALLOCATION	10,700	9,900	9,400	800	8.08%	500	Product allocation; shipments constrained
Week 7	PROMOTION	10,900	12,300	10,800	1,400	11.38%	1,500	Regional weather event surge
Week 8	PROMOTION	11,100	12,800	11,100	1,700	13.28%	1,700	Competitor price increase reaction

Segment Group	Weeks	MAE (Units)	MAPE	WAPE
Non-Promotion Weeks (1, 2, 5, 6)	4	675	6.25%	6.41%
Promotion Weeks (3, 4, 7, 8)	4	2,075	16.23%	16.31%
Total Horizon Baseline (1–8)	8	1,375	11.24%	11.83%

Interpretation & Planning Risk

Severe Systematic Under-forecasting: Demand exceeded forecast in 7 of 8 weeks ($\sum Bias = +9,400$ units). Error is heavily concentrated in promotion weeks (16.23% MAPE vs 6.25% non-promo). Unadjusted baseline forecasts create critical stockout risks for Week 10 (elevated replenishment) and Week 18 (display event).

03 & 04 DEMAND DRIVERS & UPSTREAM KPI TRACING



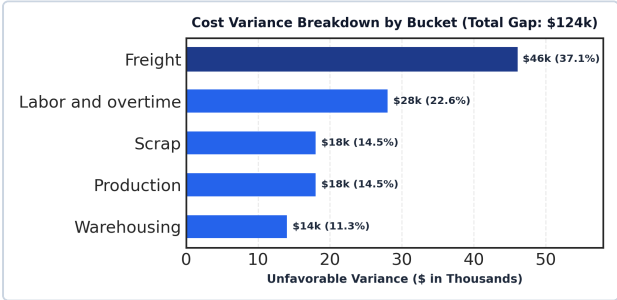
Constrained Demand Warning (Week 6)

During allocation in Week 6, shipments (9,400) fell 500 units below orders (9,900). Using shipments as a demand proxy understates true market demand and corrupts future baseline forecasting models.

KPI Metric	Target	Plant	DC	4-Wk Trend
Service Level	96%	94%	89%	DOWN
OTIF	95%	93%	88%	DOWN
Fill Rate	97%	95%	90%	DOWN
Backorders (Units)	<500	900	2,400	UP (+380%)
Days of Supply (DOS)	21 Days	18 Days	32 Days	UNEVEN
Schedule Adherence	95%	86%	91%	DOWN
Dock-to-Stock Time	24 Hrs	N/A	38 Hrs	UP (+58%)

Node Breakdown: The DC is the primary bottleneck. Despite 32 days of supply, backorders reached 2,400 units, demonstrating severe SKU mix mismatch (holding non-promo inventory while EcoClean Pods stock out).

05 & 06 COST VARIANCE & ROOT-CAUSE NARRATIVE



- Integrated Root-Cause Chain:**
- Demand Surge:** Week 3 promotion coincided with a +2,400 unit forecast error and 1,200 units of backorders.
 - Supply & Flow Breakdown:** Plant schedules collapsed (86% adherence) under late supplier packaging deliveries and packaging changeover scrap (\$36k total scrap/prod variance). DC receiving lagged at 38 hrs.
 - Service Collapse:** DC ran out of EcoClean Pods stock despite 32 days of total inventory, spiking backorders to 2,400 units and dropping service to 89%.
 - Cost Escalation:** Expedited freight (\$46k variance) and weekend overtime (\$28k variance) accounted for **59.7% of the total \$124k cost overrun**.

07 PRIORITIZED RECOMMENDATIONS FOR NEXT S&OP CYCLE

PRIORITY 1: Establish Commercial Promotional Alignment & 4-Week Lock Protocol			
Implement a mandatory 4-week promotional freeze between Sales and Retail Customers with collaborative lift modeling.			
THE EVIDENCE 16.23% promo MAPE; Week 3 promotion coincided with a +2,400 unit forecast error and 1,200 units of backorders.	EXPECTED IMPACT Reduce promo MAPE to <8%; restore DC Fill Rate to >96%; eliminate backorder surge.	COST & RISK Saves ~\$30k in spot freight/OT. <i>Risk:</i> Retailer resistance to firm lead times.	MONITORING Promo WAPE & Promo Lock Compliance (Weekly review).
PRIORITY 2: Rebalance DC Inventory Mix & Establish Fast-Pick Slotting			
Reallocate safety stock from slow SKUs to EcoClean Pods; establish dedicated fast-pick receiving lanes at DC.			
THE EVIDENCE DC DOS at 32 days while fill rate hit 90%; dock-to-stock delayed at 38 hrs.	EXPECTED IMPACT Reduce Dock-to-Stock to <24 hrs; align DOS to 21 days with correct SKU mix.	COST & RISK Saves \$14k warehousing handling & 40% DC overtime. <i>Risk:</i> Initial transfer cost.	MONITORING SKU-Level DOS & Dock-to-Stock Hours (Weekly tracking).
PRIORITY 3: Lock Dedicated Regional Lane Capacity & Enforce Supplier Packaging SLAs			
Contract dedicated freight capacity on 2 key lanes and enforce 48-hr supplier packaging delivery SLAs.			
THE EVIDENCE Freight variance +\$46k (37.1% of total gap); plant schedule adherence down to 86%.	EXPECTED IMPACT Boost Carrier Performance to >95%; raise Plant Schedule Adherence to >92%.	COST & RISK Saves \$35k in premium expedites and \$18k in plant scrap. <i>Risk:</i> Fixed carrier commitment.	MONITORING Carrier OTD % & Supplier Packaging OTD % (Weekly tracking).